BU7155- Business Data Mining
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Group 20

Student Success Analytics

Presented by

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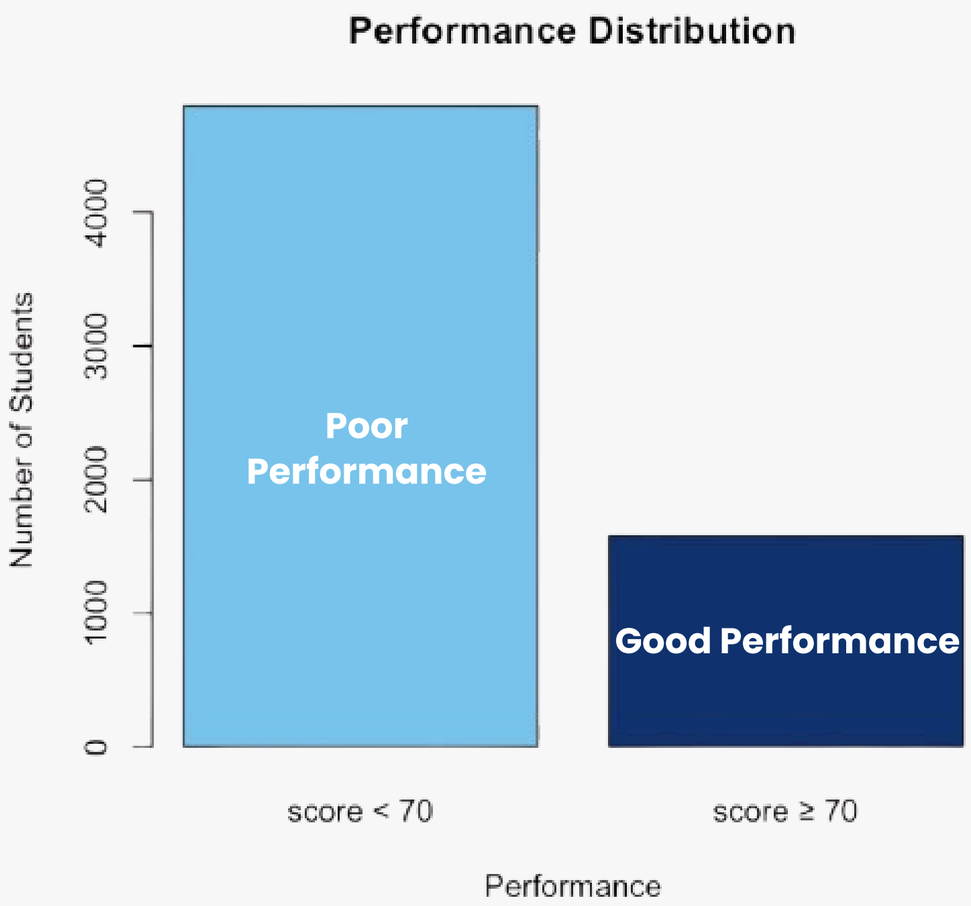
Business Problems & Data Mining Objectives

What we figured out ?

The business problem for educational institutions is that underperformance creates long-term consequences - lower results, higher drop-out rates, inefficient use of resources, and poor academic reputation.
Being able to predict which students are likely to perform poorly allows institutions to plan interventions, allocate support resources, and improve academic outcomes more effectively.

What we want to achieve ?

- Identify the correlated variables of student performance
- Evaluate and compare predictive models: Logistic Regression, Decision Tree
- Develop targeted recommendations to improve student outcomes and reduce academic underperformance



Dataset Overview

- Cleaned dataset contains **6,378 records and 19 academic + behavioural variables.**
- Created a categorical performance (result) variable for Logistic Regression.
- Set 70 as the threshold:

  ≥ 70 → Good Performance

  < 70 → Poor Performance

1. More Study Hours → Higher Scores
2. Higher Attendance → Better Performance
3. Through EDA, we found our strong Predictors:

- Hours Studied
- Attendance
- Previous Scores

4. Models used: **Logistic Regression, Decision Tree**
5. Partitioned the data into **Train(70%) & Test sets(30%).**

Project Process

Data Cleaning

1. Load dataset and remove missing values
2.Check variable types and clean formatting
3.Create binary target variable:
Result = good performance / poor performance (cutoff = 70)

Exploratory Data Analysis (EDA)

Correlation Analysis

Feature Selection

1.Select the strongest predictors based on correlation and interpretability
2.Final chosen features for modelling:
Hours_Studied, Attendance, Previous_Scores, Tutoring_Sessions

Model Development

1.Logistic Regression (interpretability)
2.Decision Tree (classification patterns)
3.Train/Test split: 70% / 30%

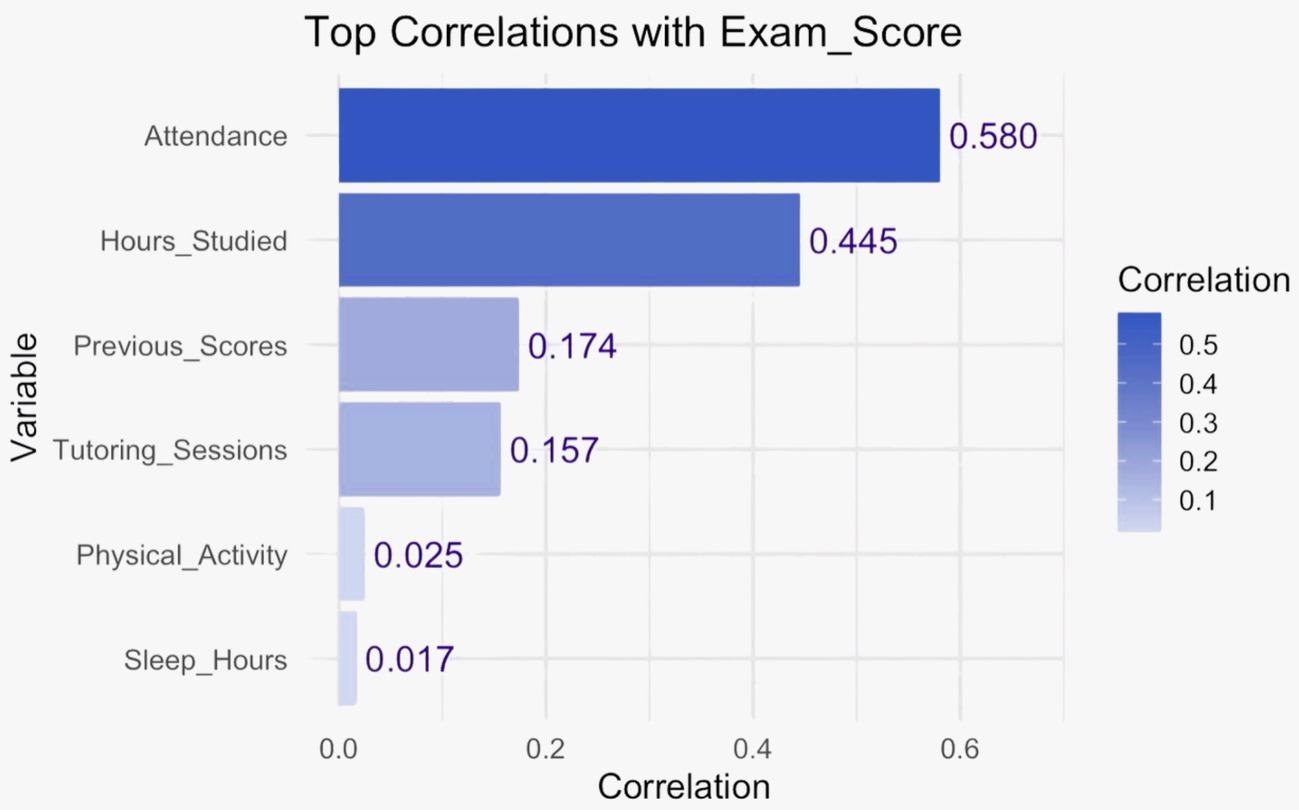
Model Evaluation

1.logistic: Confusion martrix, Accuracy, ROC curve/AUC
2.decision tree: Confusion Matrix, Accuracy and classification performance, Compare against Logistic Regression

Insights & Recommendations

1.Identify top academic risk factors
2.Detect at-risk students
3.Build strategy to improve performance

Correlation Analysis



Learning Behavior Matters Most
Attendance and study hours are the strongest predictors.

Effort > Past Ability
Current effort impacts performance more than previous scores.

Tutoring Helps, but Not Alone
Tutoring is effective only when combined with student effort.

Lifestyle Has Limited Direct Impact
Sleep and physical activity show minimal direct influence on scores.

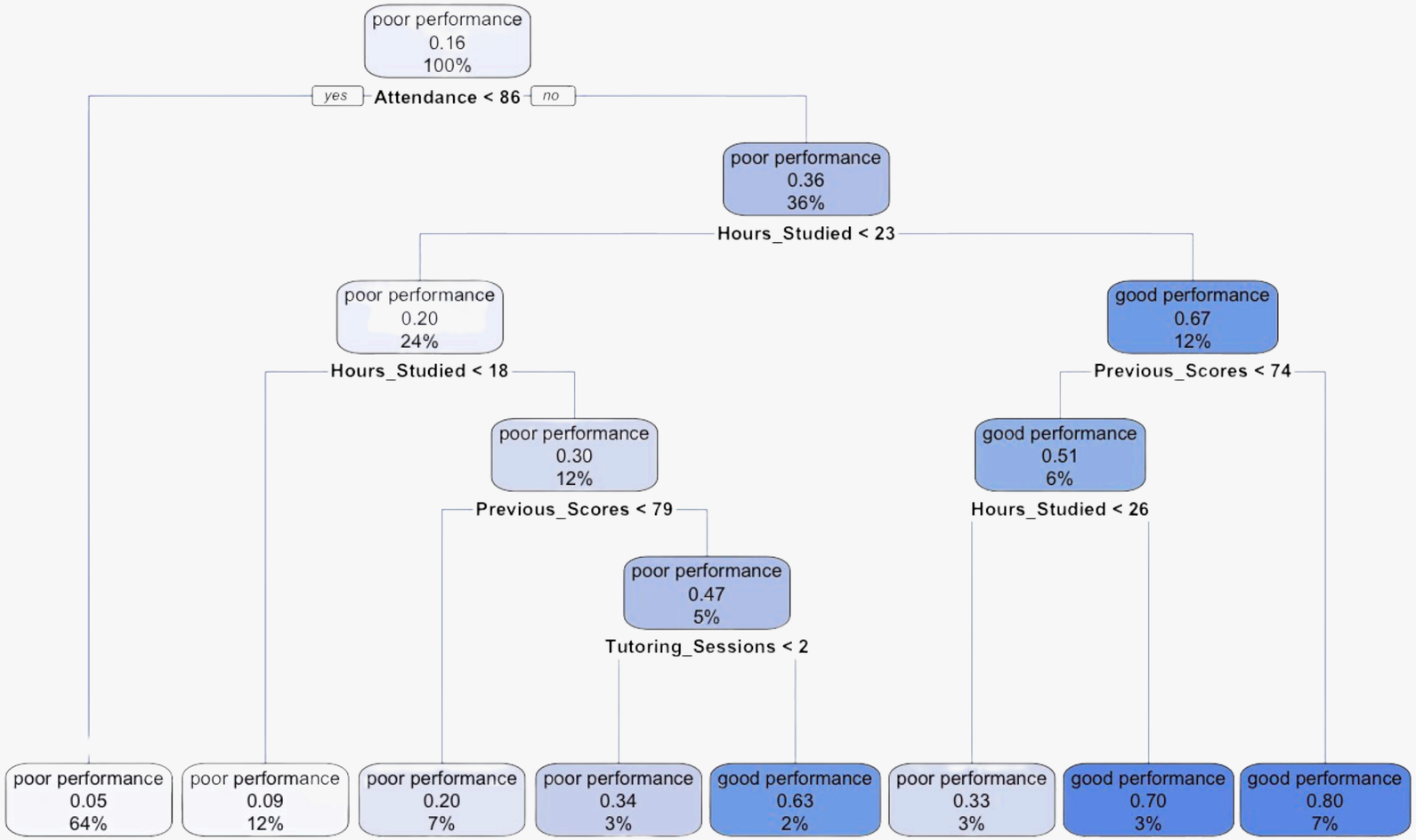
Logistic Regression

$$\text{logit}(p) = -29.665 + 0.299 * \text{Hour_studied} + 0.193 * \text{Attendance} + 0.053 * \text{Previou_scores} + 0.513 * \text{Tutorin_sessions}$$

$$p = \frac{1}{1 + e^{-(-29.665+0.299*H+0.193*A+0.053*P+0.513*T)}}$$

	Odds Ratio	Explaination
Hours_Studied	**1.348**	**1% increase in study hours, the odds increase by 34.8%**
Attendance	**1.213**	**1% increase in attendance, the odds raises by 21.3%**
Previous_Scores	**1.054**	**1% increase in previous exam score, the odds increases by 5.4%**
Tutoring_Sessions	**1.671**	**1% increase in tutoring session, the odds increases by 67.1%**

Decision Tree Results & Driver

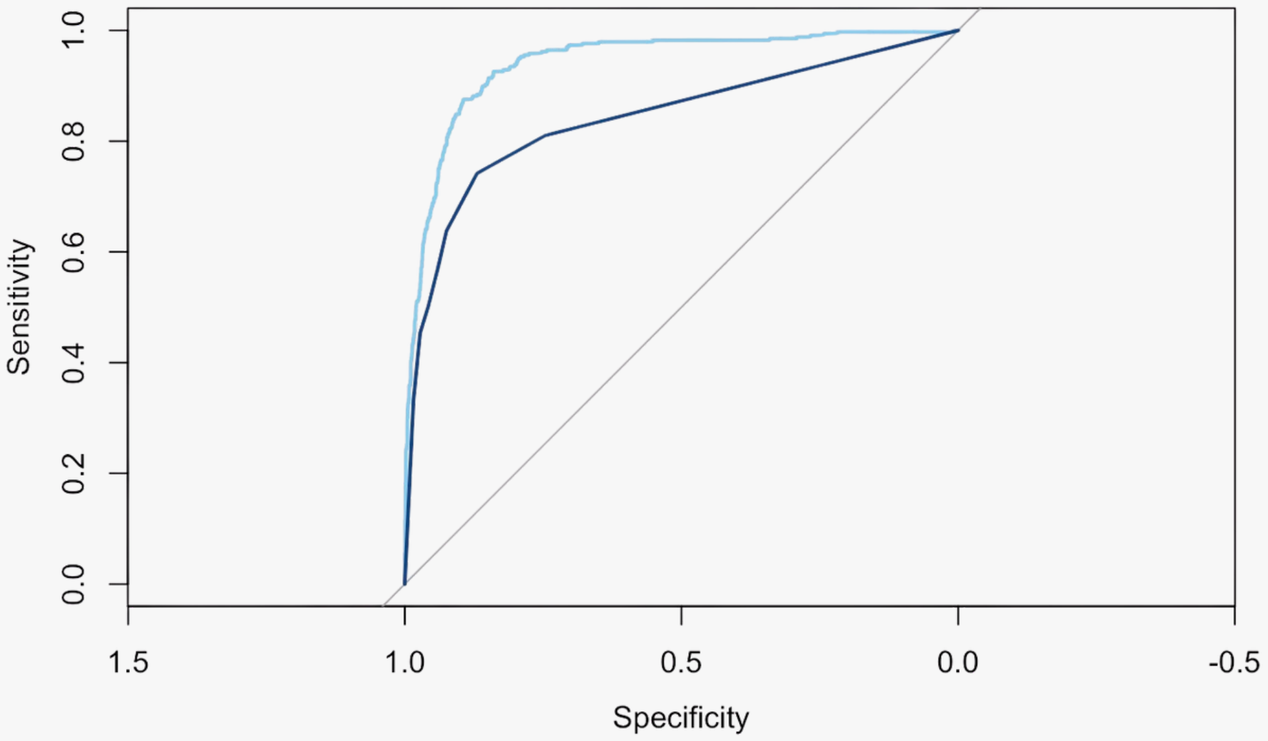


Findings:
attendance below **86%** → higher likelihood of poor performance
study hours further differentiate performance

Model Comparison

Logistic regression performs better

Metric	Logistic Regression	Decision Tree
Accuracy	90.49%	87.72%
Precision	61.42%	71.61%
Sensitivity	79.92%	50.15%
Specificity	92.15%	95.75%
AUC	Higher	Lower



Action Plan

Fix Attendance First
- Alerts for <85% attendance
- Immediate follow-up

Strengthen Study Routines
- Weekly study plans
- Study-skills workshops

Target Tutoring
- Prioritize Poor Performance students
- Link tutoring to course content

Promote Effective Behaviors
- Encourage consistent hours + participation
- Shift student mindset: effort > talent

Use Risk Levels to Allocate Support
- High-risk: close monitoring + intervention
- Medium-risk: tutoring + study planning
- Low-risk: maintain habits

Appendix 1

https://github.com/nisawooo/student-success-analytics

1. Model Description

Two supervised learning models were used to classify students into good performance and poor performance categories:

- Logistic Regression – a probabilistic, interpretable classification model
- Decision Tree – a rule-based, interpretable model

Target variable: Result (good performance vs poor performance)

Key predictors: Attendance, Hours_Studied, Previous_Scores, Tutoring_Sessions

2. Intended Use

The model is intended for academic performance risk prediction, to help identify students who may require additional support, such as tutoring or attendance intervention.

Not intended for high-stakes decision-making.

Appendix2

3. Fairness Considerations

- Gender Bias: No strong evidence of gender-based prediction bias. Model coefficients showed minimal gender effect.
- Socioeconomic or parental involvement bias: Some influence exists (Parental_Involvement), but the impact is relatively small compared to Attendance and Hours Studied.
- Attendance and study hours dominate predictions: These are behavioural rather than demographic factors, reducing fairness concerns.
- Class imbalance handled: Accuracy interpreted alongside sensitivity and specificity to avoid misleading conclusions.

4. Limitations

- Does not establish causal relationships.
- Decision Tree may be unstable and sensitive to sampling.
- Performance relies heavily on attendance distribution in the dataset.
- The model may not generalise to significantly different student populations.

Thank You So Much!

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